

Label smoothing

This report briefly explains about label smoothing & the areas, how it should be used

The Reason for the Paper

Before label smoothing people used hard labels in neural network

Hard labels:-

They are simply outputs of neural network. But the problem here is these outputs are chosen way too confidently by the system with considering other option.

ie if you provide a blurry image of a dog. Well, according to statistical analysis it is confirmed that it can't be at max 80% accurate it is a dog but the neural network will confidently say it is 100% a dog. This phenomenon is called overconfidence.

In this paper we are going to provide proper evidences and test results to ^{solve} the existing problems created by the ~~exist~~ overconfident existing neural network with label smoothing

Another major problem is ~~to~~ the proper usage of label smoothing.

Problems created by overconfident neural systems:

- From the example I provided above you can see it can be only 80% chance it is a dog. But, the neural network will predict it as 100% dog.
- Sometimes, the photo may turn out to be cat as there is still 20% chance ~~as~~ per statistics for it to be any other animal. which significantly affects the accuracy of our model.

Label smoothing:-

• Its primary use is to prevent the system from getting overconfident.

• It is a normal probabilistic which goes like

$$y_K^{LS} = (1-d)y_K + d/K$$

where

y_K = original hard label

y_K^{LS} = smoothed target label

K = number of classes

• Function of label smoothing

• It assigns high ~~percentage~~ probability value for the predicted correct outcome and instead of leaving remaining output it assigns the remaining probability value equally distributed to them.

• How it changes the model.

Before Label

smoothing

System response for a image inserted

it is 100% a dog

After Label

Smoothing

System response for a image is it is mostly a dog but it is also maybe a cat, cow etc.

Advantages of Label Smoothing

(i) Improves accuracy

(ii) Reduces overconfidence

(iii) Improves calibration of probabilities

(iv) Easy to implement

(v) Reduces overfitting

Disadvantages of Label Smoothing

(i) Doesn't work well in knowledge distillation

(ii) Improves the accuracy of teacher while heavily reducing accuracy of student model.

(iii) Converts sharp outputs to a partially mixed one.

Possible future improvements:

(i) To improve label smoothing to work well for training distillation.

(ii) Making label smoothing to work on data which is not clear

(iii) Making label smoothing adapt to different type of data on its own.